PULSETWIN / DIGITAL TWIN CHALLENGE 2026

Personal health, simulated.

01 A personal blood-pressure model that updates with observations.

02 Explore scenarios. Inspect uncertainty. Export the evidence.

03 Open-source research prototype • Synthetic demonstration

01 / THE PROBLEM

A chart tells only half the story.

01 Daily readings are noisy and disconnected from daily habits.

02 Generic averages hide individual patterns.

03 A useful simulation must expose its assumptions.

02 / SCOPE AND USERS

One focused healthcare use case.

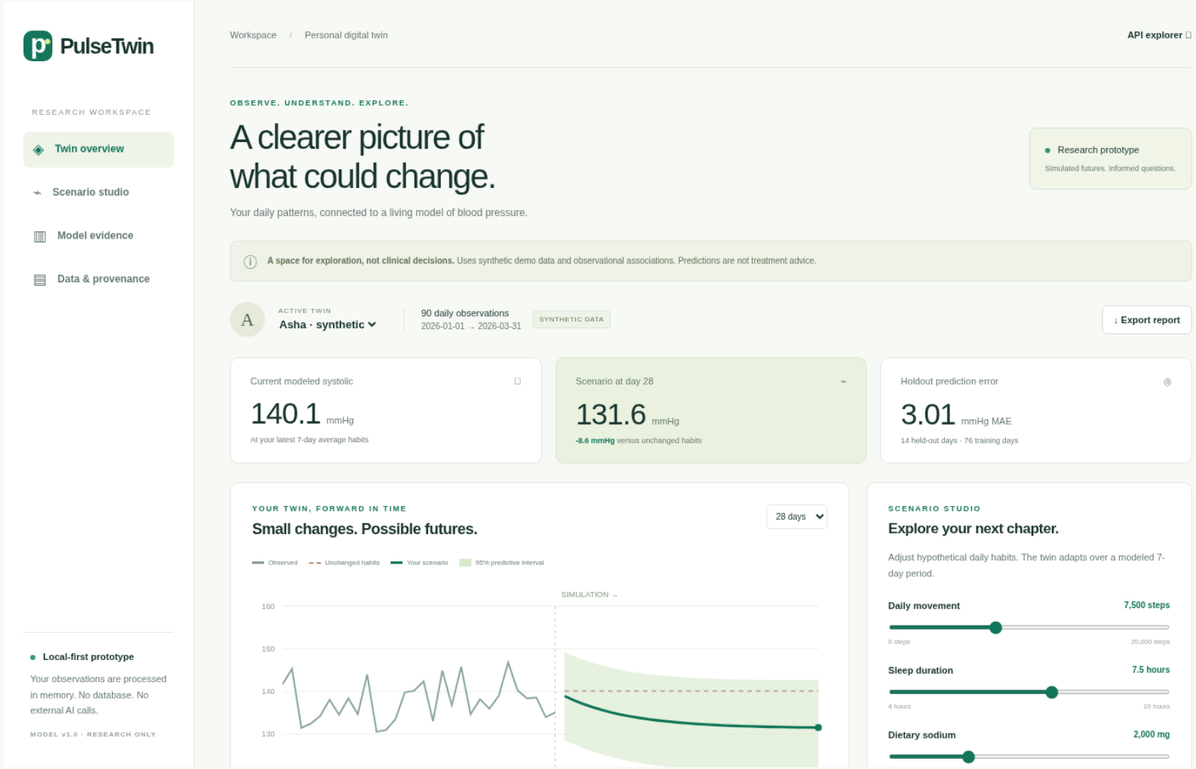
01 Longitudinal cardiometabolic research and education.

02 Systolic blood pressure as the modeled outcome.

03 No diagnosis, medication advice or autonomous decisions.

03 / DASHBOARD WALKTHROUGH

A working research workspace.



01

Three synthetic profiles, each with 90 daily observations.

02

Current modeled SBP, scenario outcome and holdout error.

03

Visible provenance and research-only boundaries.

04 / OBSERVATION CONTRACT

Daily data becomes a personal state.

01 Date, SBP, steps, sleep, sodium and stress.

02 28–730 consecutive days; no silent gap filling.

03 JSON API and CSV import share the same schema.

05 / BAYESIAN PERSONALIZATION

A model you can inspect.

01 Four standardized habit features plus a personal intercept.

02 Gaussian priors stabilize small histories.

03 Posterior coefficients and covariance use closed-form NumPy algebra.

06 / THE LATENT STATE

Updating the twin over time.

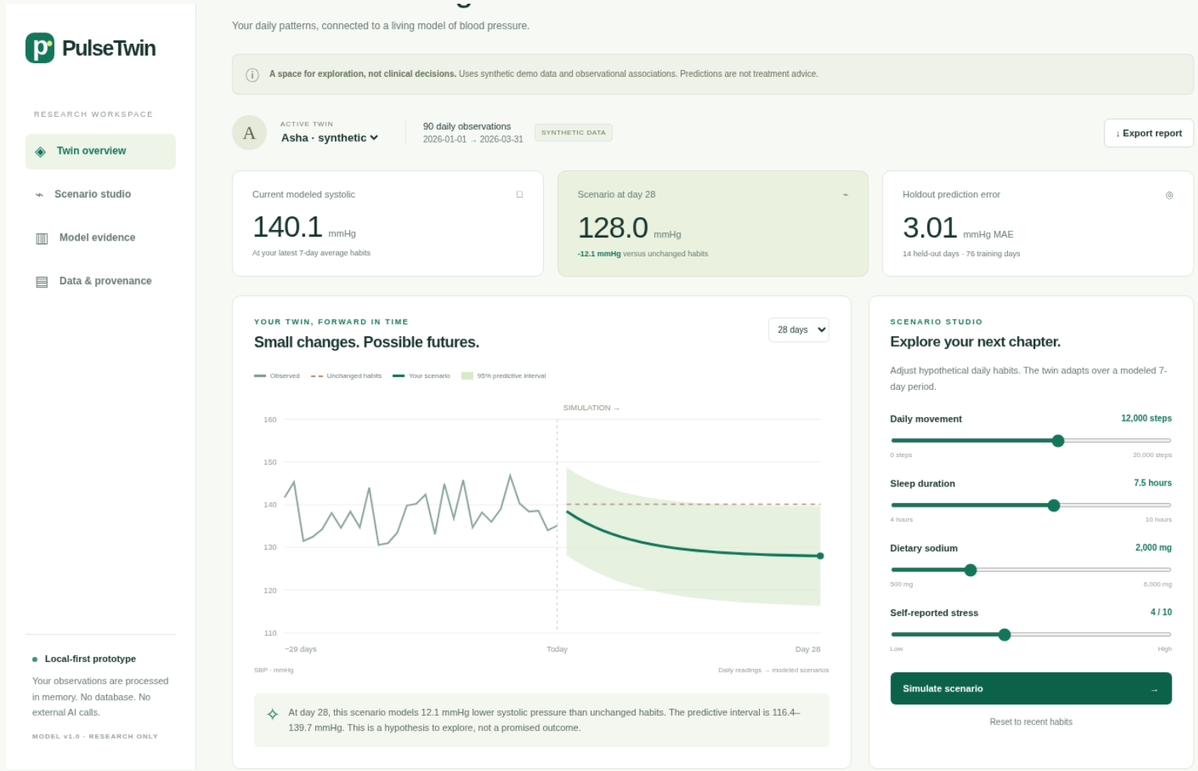
01 Daily residuals update a scalar Kalman state.

02 New observations change the personal fit and current state.

03 Human-in-the-loop prototype; no live device synchronization.

07 / SCENARIO STUDIO

Explore a conditional future.



01

Adjust movement, sleep, sodium and self-reported stress.

02

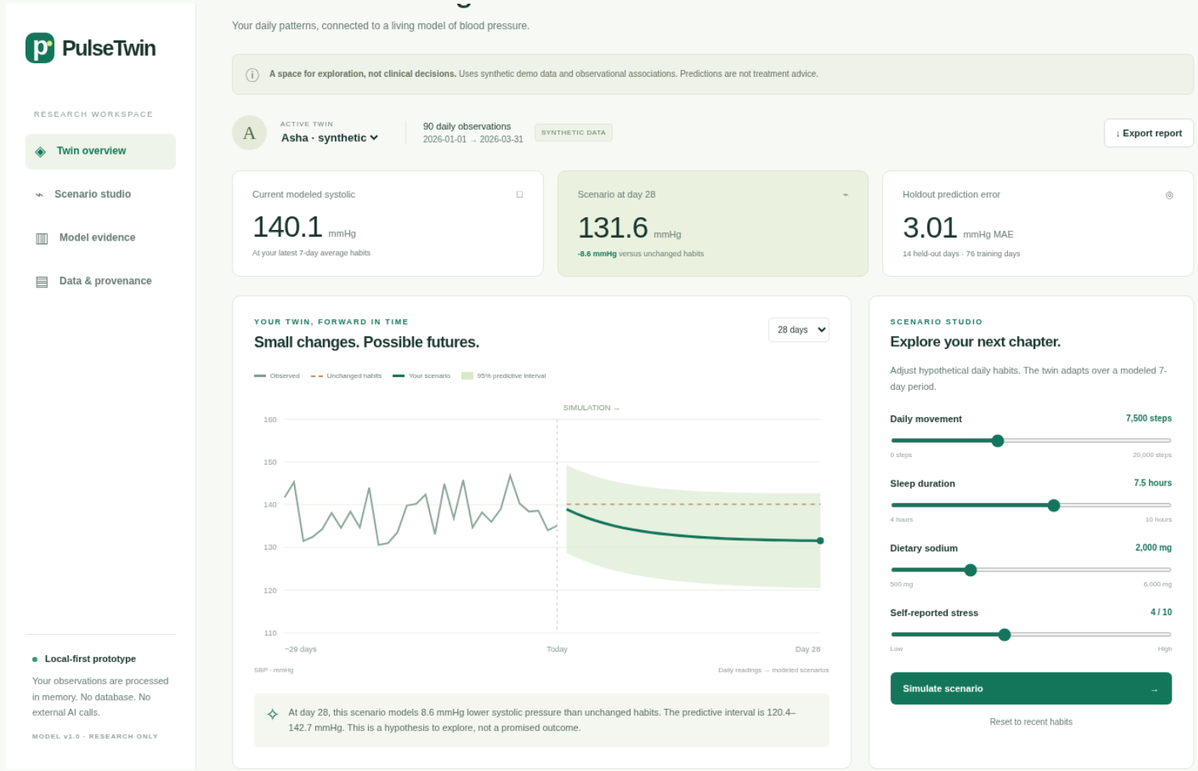
Choose 14, 28, 56 or 90 days in the interface.

03

Changes approach the target with an assumed seven-day time constant.

08 / PREDICTIVE UNCERTAINTY

Read the band, not just the line.



01

Observed history, unchanged-habits baseline and scenario curve.

02

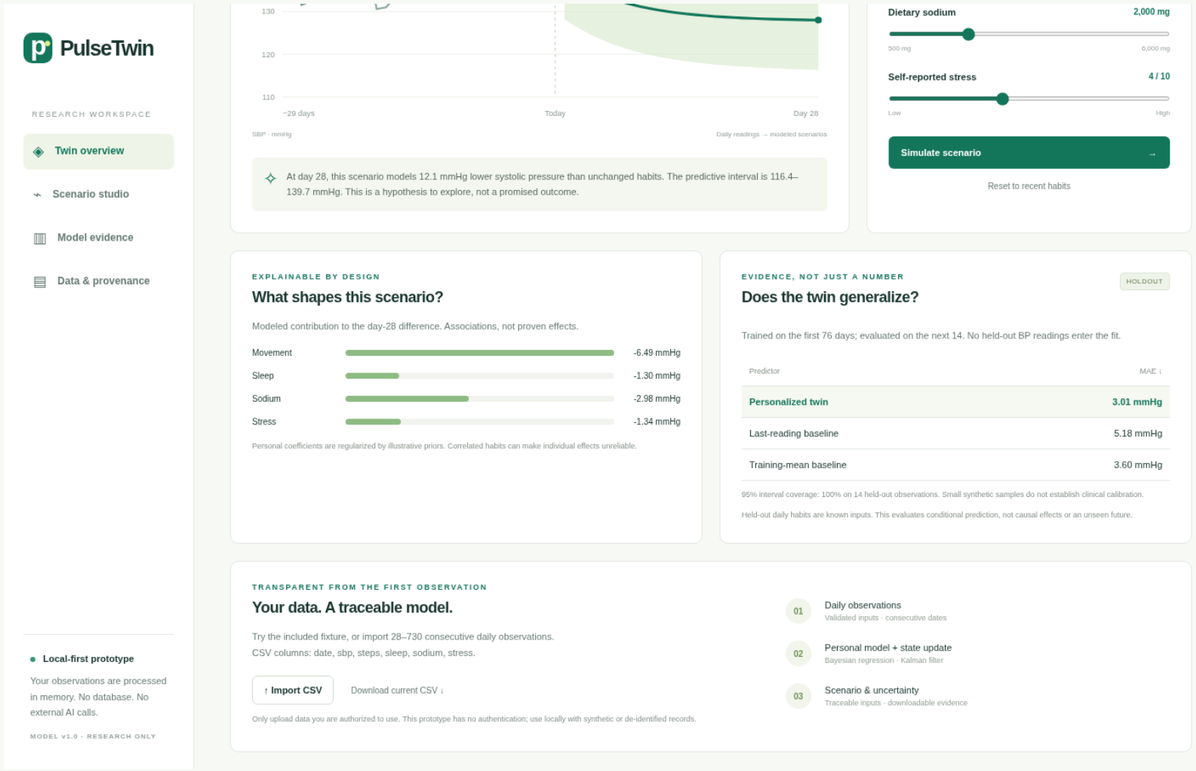
95% Gaussian posterior predictive interval.

03

Model uncertainty is narrower than real-world uncertainty.

09 / CONTRIBUTION BREAKDOWN

Explain the difference.



01

Feature contributions sum to the modeled scenario difference.

02

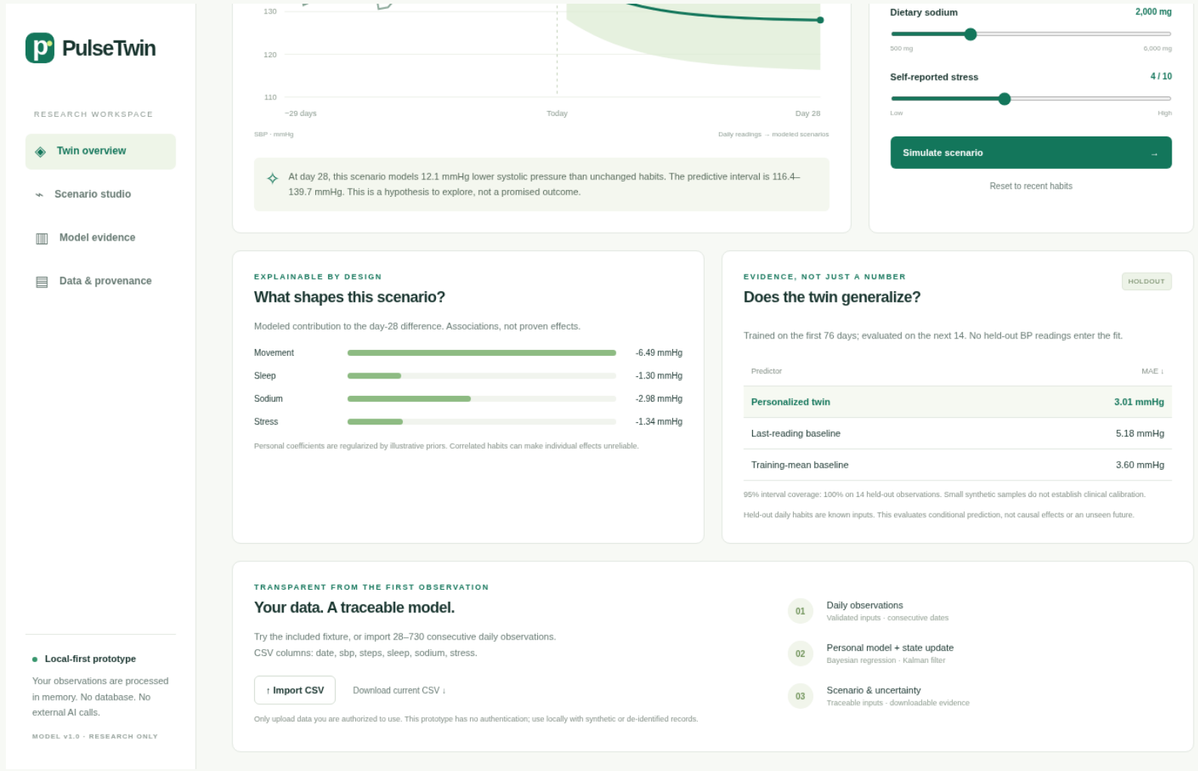
Signs come from each personal fit.

03

Correlated predictors limit causal interpretation.

10 / TEMPORAL VALIDATION

Test against simpler answers.



01

First 76 days train; final 14 days test for each demo profile.

02

No held-out BP labels enter the fit or state update.

03

Compare with last-reading and training-mean predictors.

11 / RESULTS WITH LIMITS

Measured prototype outcomes.

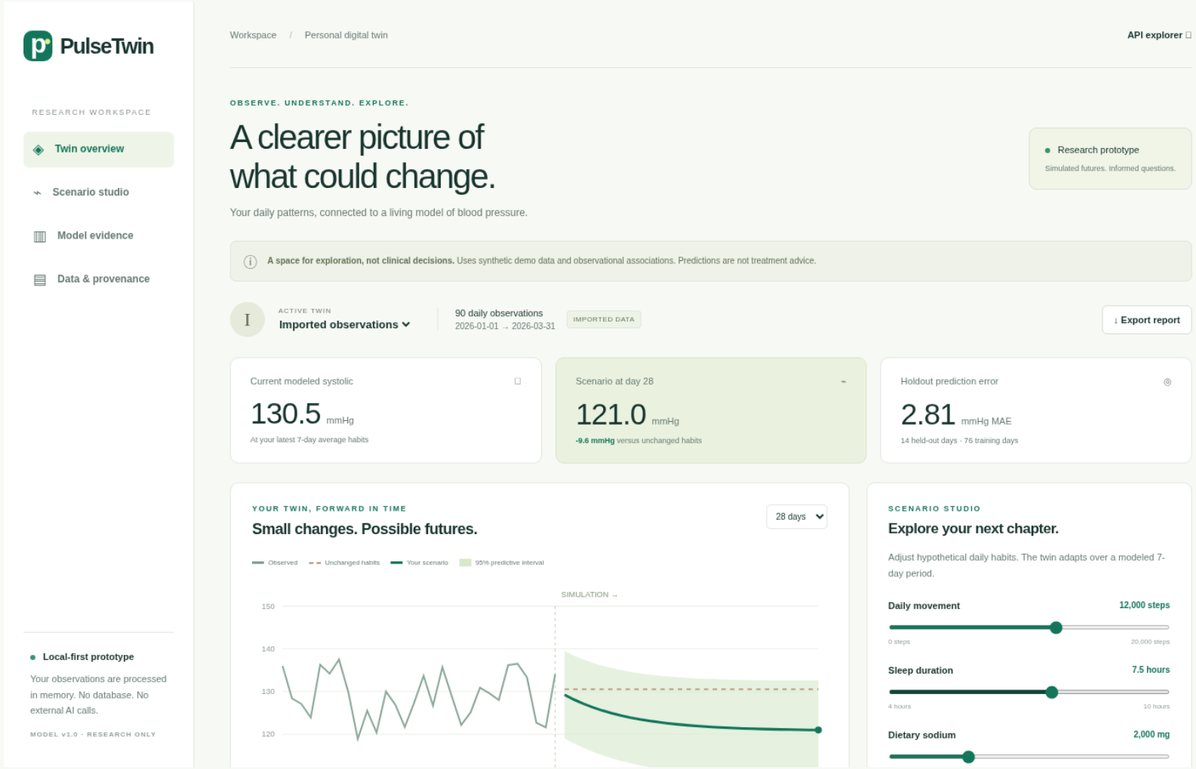
01 Twin MAE: 3.01 / 3.19 / 2.81 mmHg across the three fixtures.

02 Training-mean MAE: 3.60 / 3.85 / 4.47 mmHg.

03 Synthetic data only; 14 test observations per profile.

12 / IMPORT AND EXPORT

Try your own authorized fixture.



01

Import a valid CSV and see the provenance badge change.

02

Switch among synthetic and imported observations.

03

Export raw inputs, scenario, forecasts and validation as JSON.

13 / SYSTEM DESIGN

Simple architecture. Real endpoints.

01 Browser UI FastAPI validation NumPy model.

02 Stateless requests; no database or external AI calls.

03 Reproducible fixtures, evaluation and local artifacts.

14 / ENGINEERING EVIDENCE

Validation beyond a screenshot.

01 14 model and API tests pass.

02 Real Chromium checks cover imports, exports and mobile layout.

03 Leakage, non-finite inputs and uncertainty behavior are tested.

15 / PRIVACY AND SAFETY

Boundaries are part of the product.

01 Synthetic fixtures; no credentials or patient records included.

02 Observations stay in browser and request memory.

03 No diagnosis, treatment recommendation or clinical certification.

16 / ROADMAP

A credible path beyond the prototype.

01 Consented longitudinal data and external evaluation.

02 Calibration, missingness, subgroup analysis and clinician review.

03 Secure device ingestion before any prospective deployment.

17 / REPRODUCTION AND HANDOFF

Open the code. Reproduce the evidence.

01 Python 3.12 · FastAPI · NumPy · vanilla JavaScript.

02 README, MIT license, model card, PDFs and test commands.

03 Personal details and authorized YouTube publication remain external steps.